SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	29 June 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction System
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file or (.env.example similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes

10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

```
rising_waters/
├── data/
│   ├── generate_data.py      # builds a synthetic dataset (optional/demo)
│   └── flood_data.csv      # historical weather dataset (real or synthetic) ├──
model/
│   ├── train.py            # trains & compares all 4 classifiers
│   ├── best_model.pkl      # saved best-performing model (generated)
│   ├── scaler.pkl          # saved StandardScaler (generated)
│   └── results.json        # accuracy/precision/recall/F1 per model (generated)
├── templates/
│   ├── base.html
│   ├── index.html          # prediction form + result
│   └── about.html          # model performance comparison page
├── app.py                # Flask application
├── requirements.txt └──
README.md
```

Step 3: Access Details

Field	**Details**
Login Credentials (Demo)	N/A (No login authentication implemented)
Platform / Hosting Provider	Localhost (Current),Render.
Repository Link	https://github.com/lasya365/AI-ML-and-GEN-AITrack-Project-Template.git
Demo Video Link	https://drive.google.com/file/d/1zBAteBe1dImXZmlTK6vwHE2Brlp6mhb/view?usp=sharing

Step 4: Run Instructions

- Clone or download the project source code.
- Open the project folder in **VS Code**.
- Open a terminal inside the project folder.
- Install the required dependencies:

```
pip install -r requirements.txt
```

- Ensure the following files are present:

```
rising_waters/data/generate_data.py
rising_waters/model/ train.py
rising_waters/model/best_model.pkl
rising_waters/templates/base.html
rising_waters/ templates/ index.html
rising_waters/  templates/about.html
rising_waters/app.py
```

- Run the Flask application:

```
python app.py
```

- Open a web browser and visit:

```
http://127.0.0.1:5000/
```

- Enter the required weather parameters.
- Click **Predict** to view the flood prediction, probability, and risk level.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Predictions are based on historical dataset only.	Future work: Integrate real-time weather APIs.
2	The application does not display flood maps or geographical visualization.	Future enhancement: Integrate Leaflet, Folium, or Google Maps API.
3	The application currently runs on localhost only.	Future deployment on Render, IBM Cloud, or other cloud platforms.